

IBM Data Analyst Capstone Project

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Date:22/05/2026



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OUTLINE



- Executive Summary
- Introduction
- Methodology
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 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- Collected and analysed Stack Overflow Developer Survey data
 - python
 - pandas
 - POWER BI
- Cleaned and processed the dataset using Python and pandas
- Built dashboards to present insights effectively.



INTRODUCTION



- The dataset contains responses from developers worldwide
- The analysis focuses on programming languages, salaries, and work preferences
- Python, Pandas, and visualization tools were used for analysis.
- The goal is to identify trends and insights from developer data.
 - Data Cleaning
 - Dashboard Creation



METHODOLOGY



- Data collection from survey dataset
- Data cleaning using Pandas
- Data analysis using Python
- Data visualization using charts
 - Python
 - Power BI / Visualization Tools



RESULTS

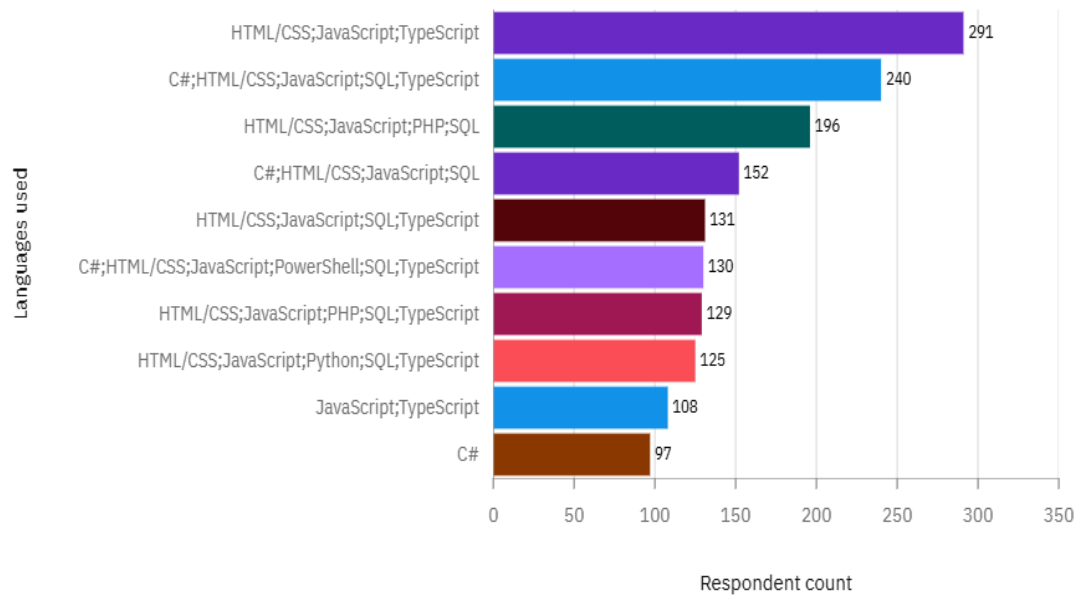
- Analysis identified popular programming languages
- Developer preferences were analyzed
- Charts and dashboards improved understanding of data



PROGRAMMING LANGUAGE TRENDS

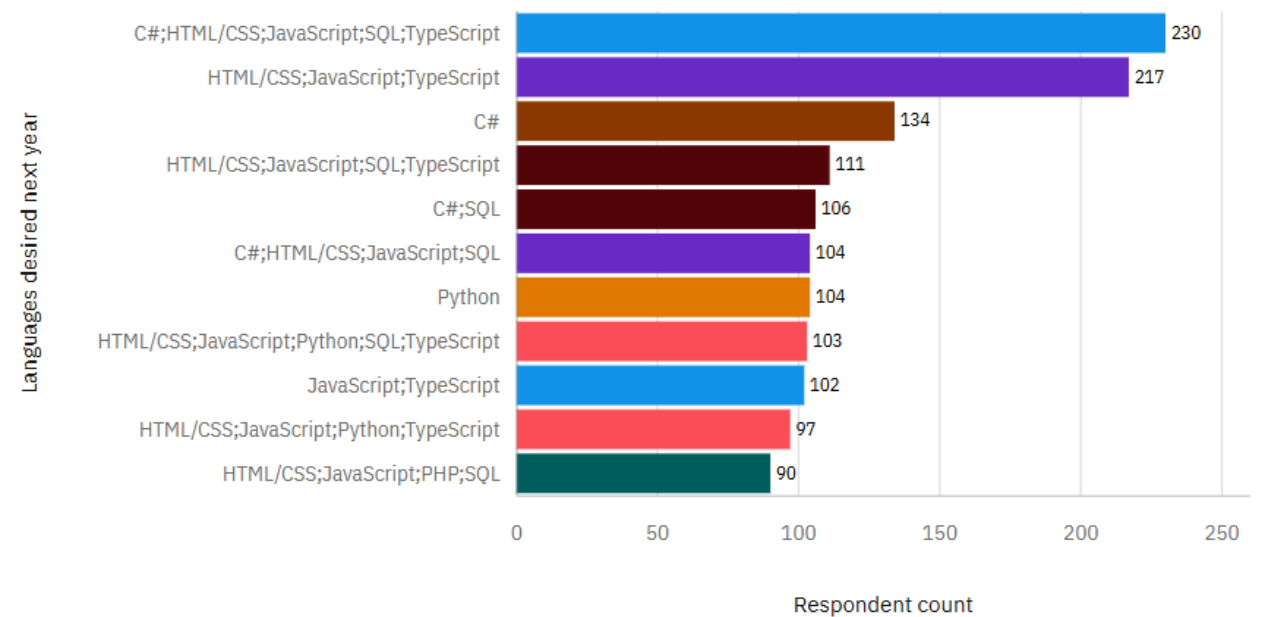
Current Year

Top 10 Programming Languages Used



Next Year

Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript and Python are among the most popular languages
- HTML/CSS and SQL are widely used technologies
- Developers continue to use multiple programming languages

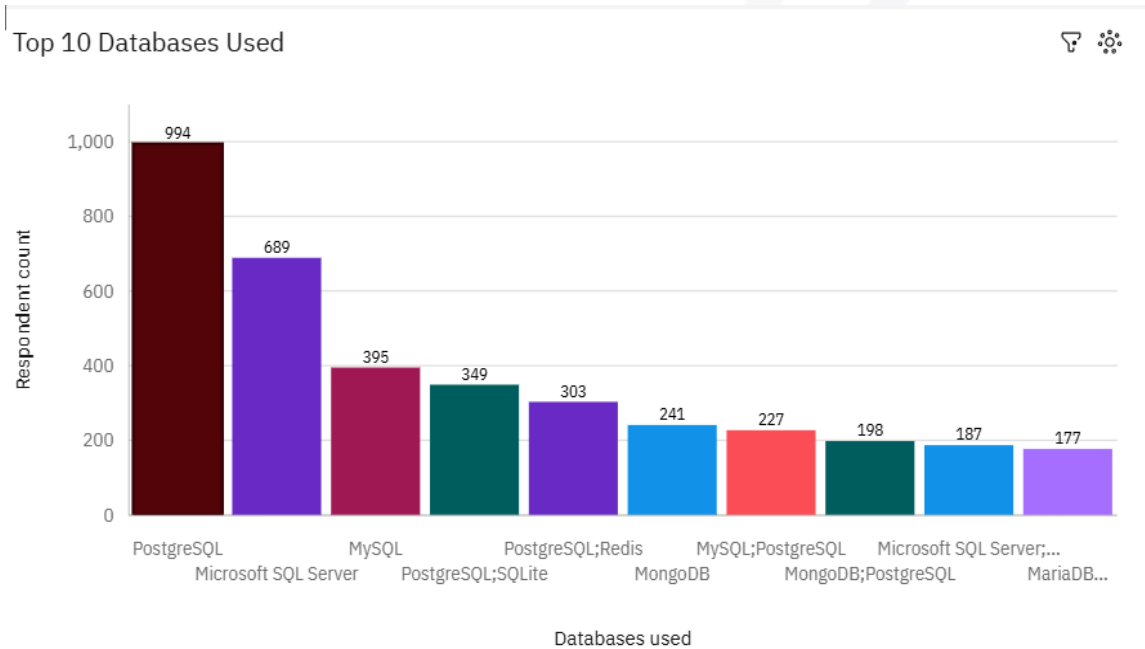
Implications

- Learning Python and JavaScript improves career opportunities
- Full-stack development skills are in high demand
- Database knowledge is important for developers

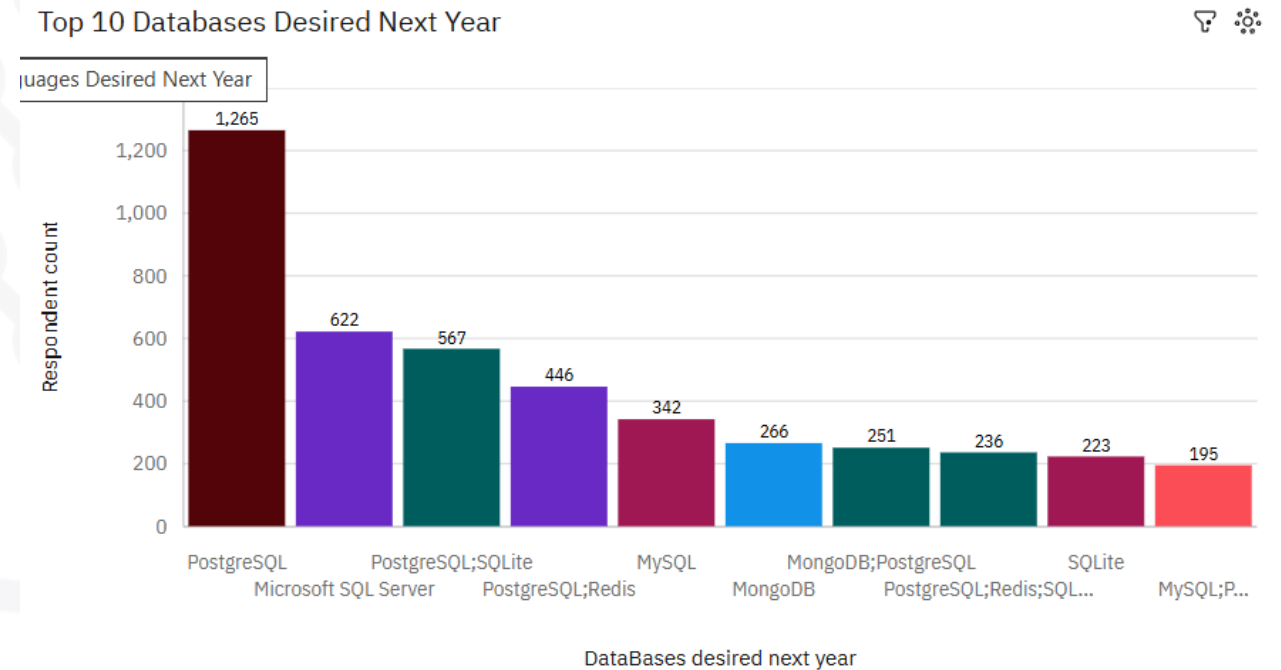


DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL and MySQL are widely used databases
- MongoDB is popular for modern applications
- Microsoft SQL Server remains important in industry

Implications

- Database management skills are highly valuable
- SQL knowledge is essential for data-related roles
- Cloud and NoSQL databases are growing rapidly



DASHBOARD

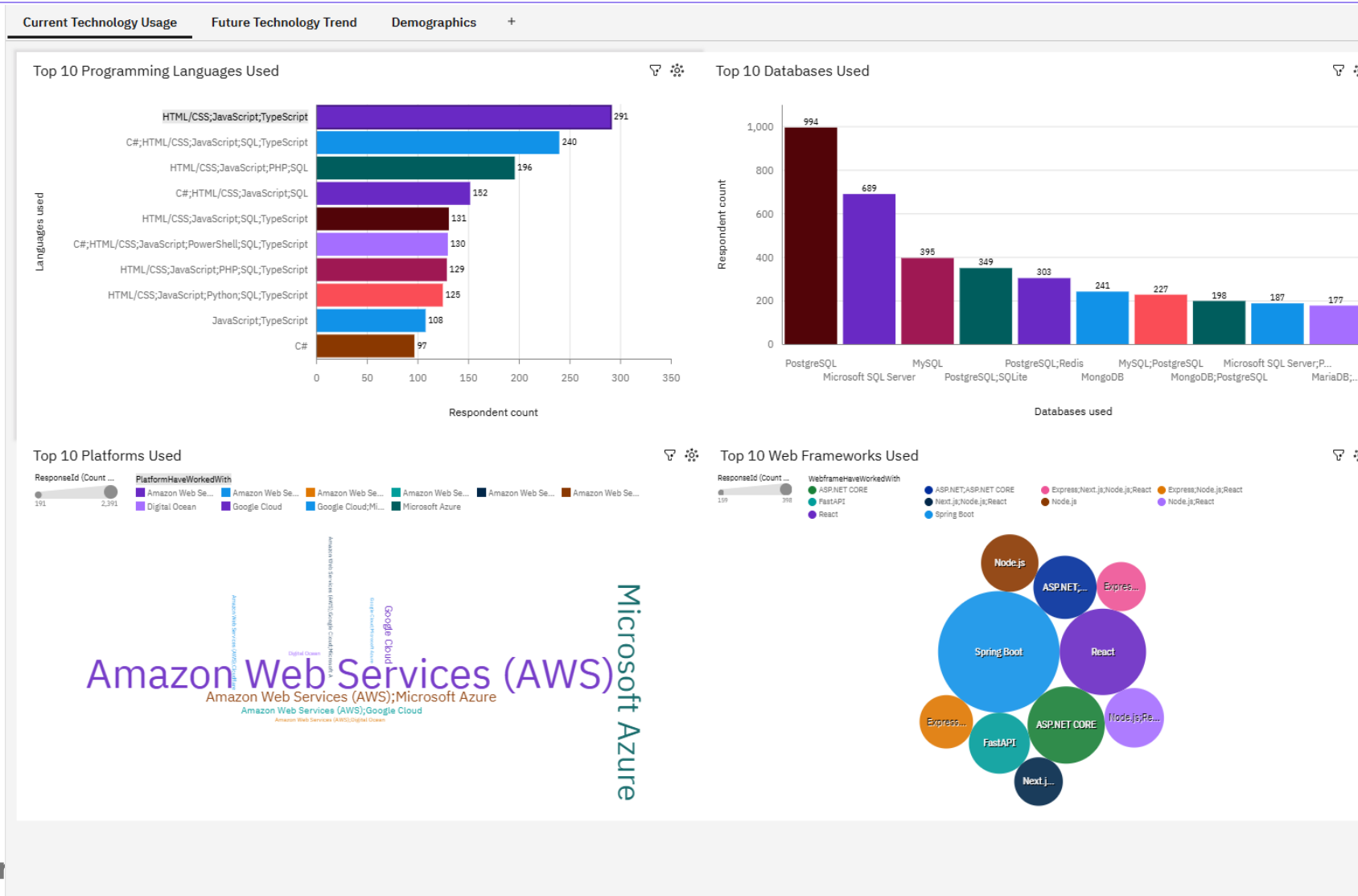


IBM Cognos Analytics Dashboard

Dashboard includes 3 tabs: Current Technology Usage, Future Technology Trend, and Demographics.

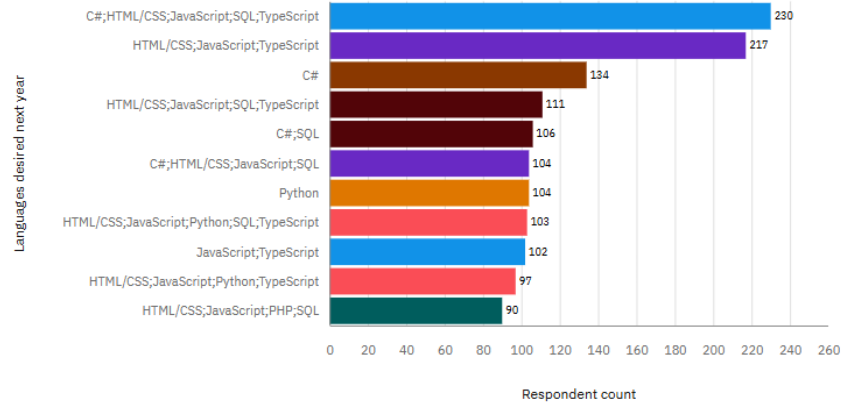


DASHBOARD TAB 1

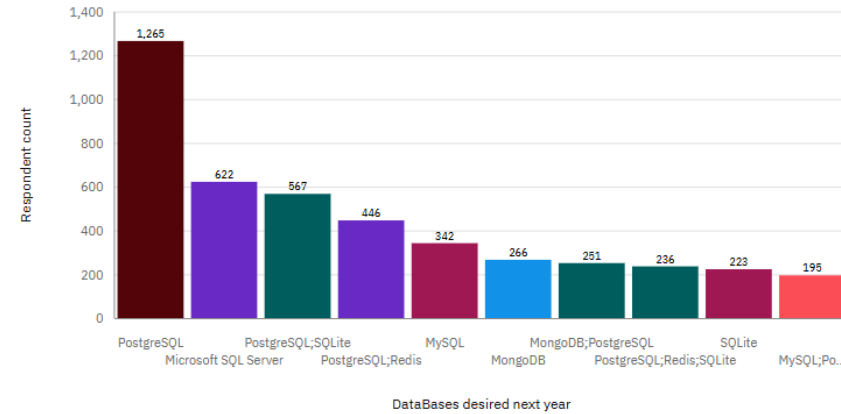


DASHBOARD TAB 2

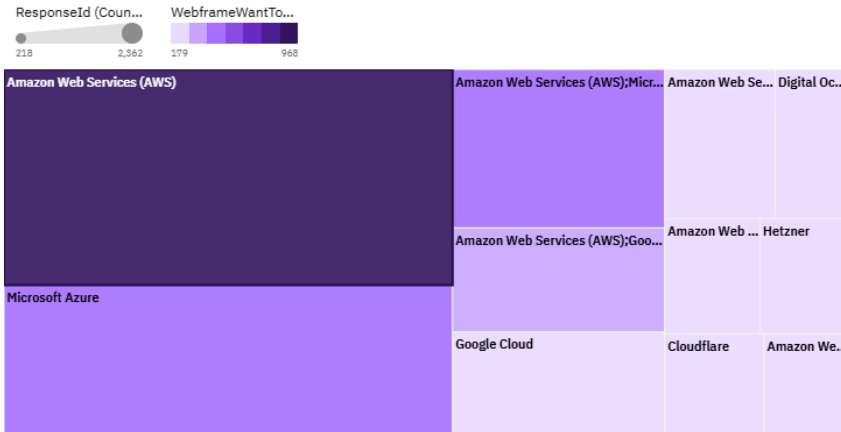
Top 10 Languages Desired Next Year



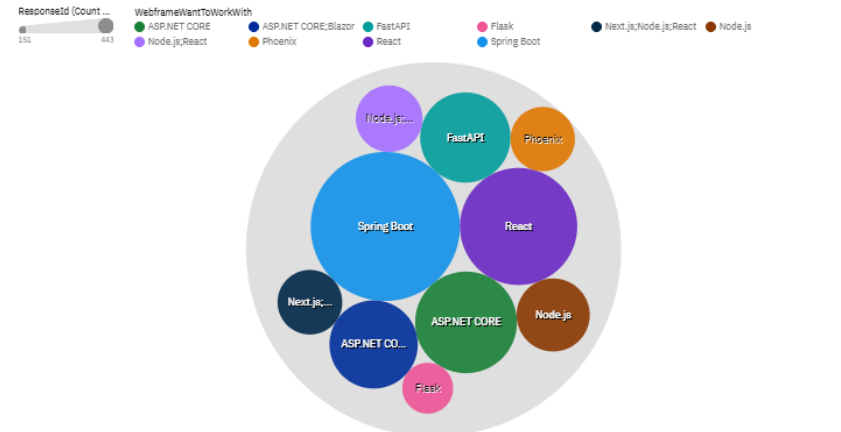
Top 10 Databases Desired Next Year



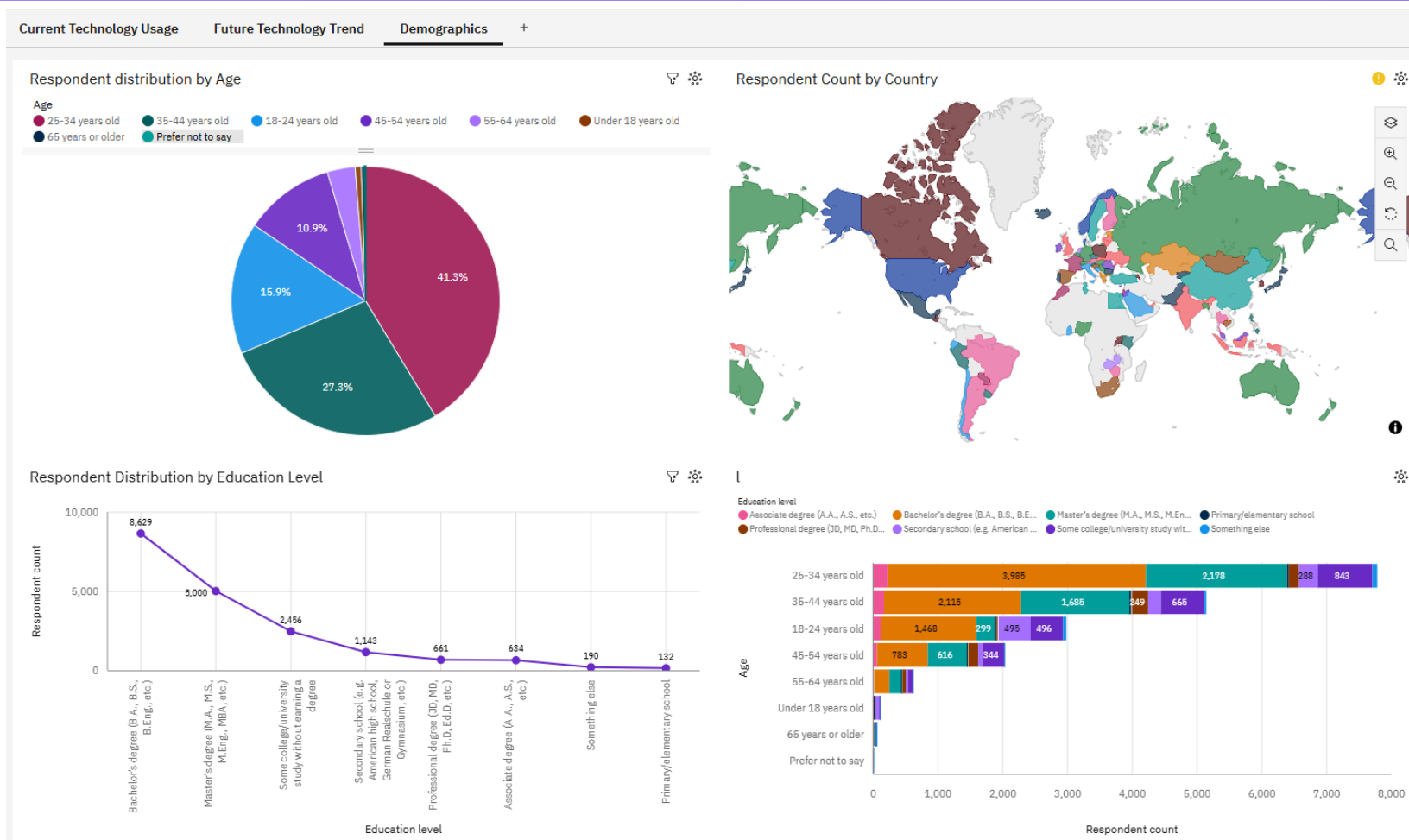
Top 10 Desired Platforms



Top 10 Desired Web Frameworks



DASHBOARD TAB 3



DISCUSSION



- Python and JavaScript are among the most popular technologies.
- Database technologies remain important in software development.
- Cloud platforms such as AWS are widely used. Dashboards improve analysis and decision-making process.



OVERALL FINDINGS & IMPLICATIONS

Findings

- Python and JavaScript are the most popular programming languages.
- PostgreSQL and MySQL are widely used databases.
- AWS and cloud technologies are growing rapidly.

Implication

- Learning Python and JavaScript improves career opportunities.
- Database knowledge is important for data-related roles.
- Cloud computing skills are becoming essential in industry.

CONCLUSION



- Python and JavaScript are leading technologies.
- PostgreSQL and MySQL are highly preferred databases.
- Cloud technologies like AWS are growing rapidly.
- Data analytics helps in better business decision-making.

APPENDIX

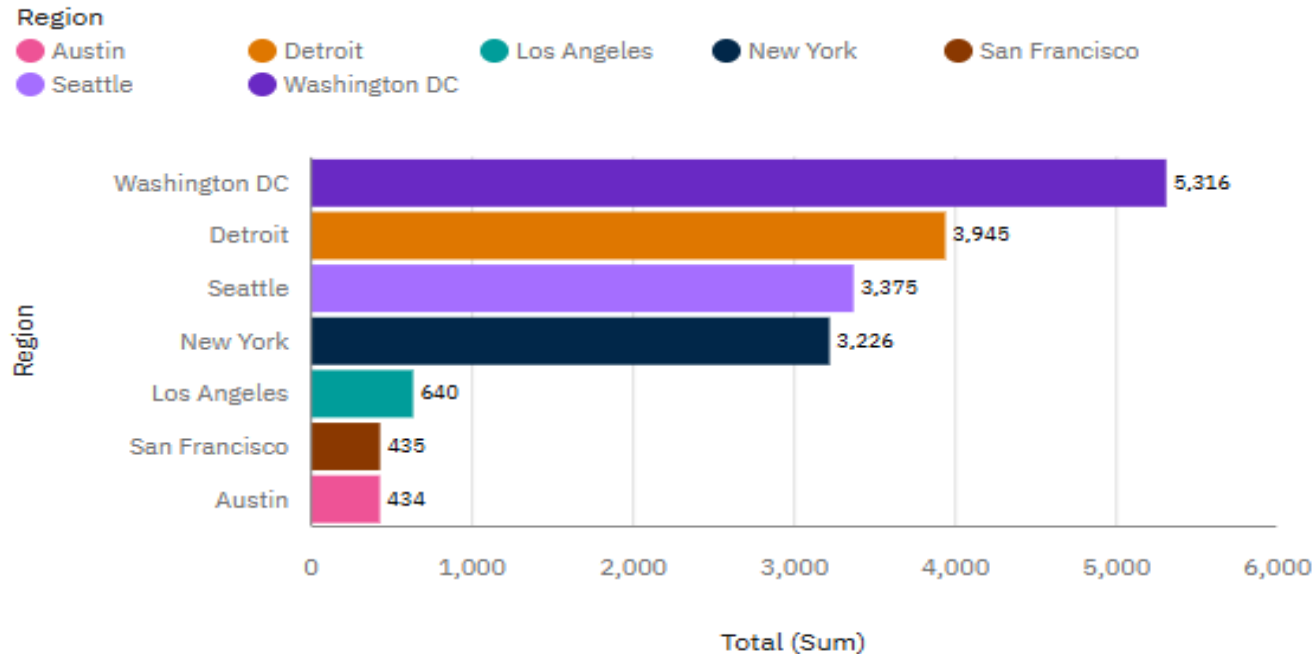


- Include any relevant additional charts, or tables that you may have created during the analysis phase.
- Additional dashboard screenshots included.
- Dataset used: Stack Overflow Developer Survey.
- Tools used: Python, Pandas, Cognos Analytics.



JOB POSTINGS

Total by Region colored by Region

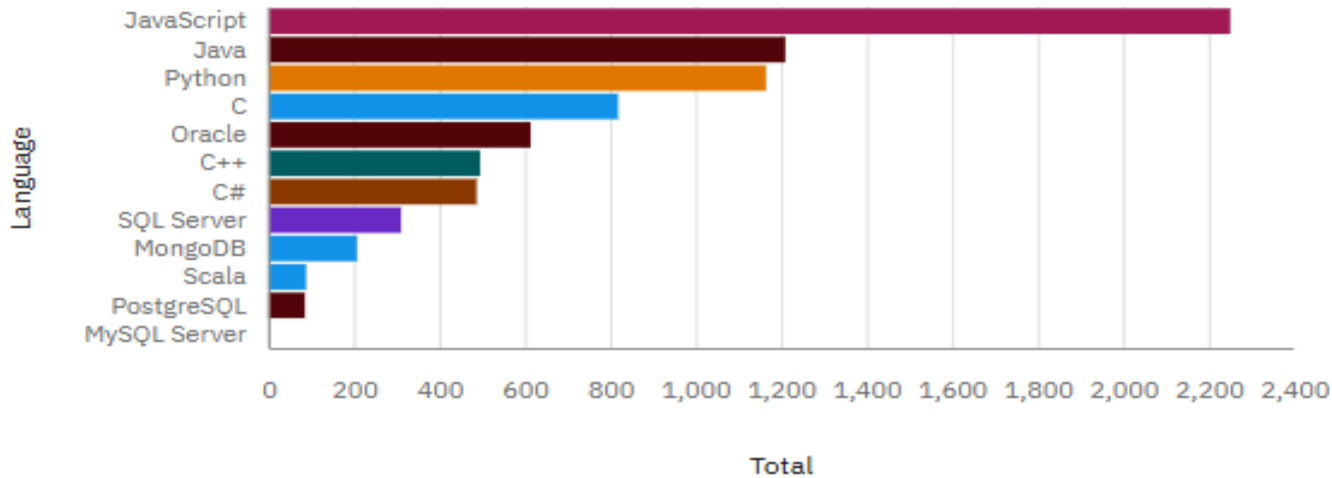
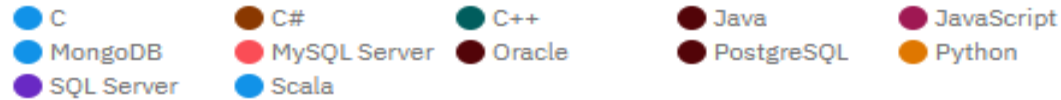


- Washington DC and Detroit have the highest number of job postings.
- Tech jobs are concentrated in major cities.

POPULAR LANGUAGES

Popular Languages

Technology



- JavaScript and Python are the most popular programming languages.
- SQL and HTML/CSS are widely used by developers.
- Multi-language programming skills are in high demand.

